

MINI PROJECT REPORT

Data Science | Academic Year 2025–2026

Predict & Explain: Identifying Heart Disease Risk Factors

Using Patient Clinical Data and Machine Learning

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SDG Goal: 3 – Good Health and Well-being | Domain: Healthcare / Bio-Informatics

1. Introduction

Heart disease is a leading cause of preventable death worldwide. Early prediction using routinely collected clinical data can significantly improve patient outcomes by enabling timely medical intervention. This project develops a machine learning pipeline that ingests 13 clinical features and outputs a prediction of heart disease presence, achieving reliable accuracy without invasive procedures.

The project follows a complete Data Science workflow: data collection, preprocessing, exploratory analysis, model development, evaluation, and result interpretation. Five classification models are benchmarked, and the best performer is identified using F1-Score as the primary metric.

1.1 Objectives

- Build a binary classification model for heart disease prediction from clinical data.
- Perform comprehensive Exploratory Data Analysis to understand feature distributions and correlations.
- Compare five machine learning algorithms across five evaluation metrics.
- Visualise feature importance and model performance through plots.
- Align with SDG Goal 3: Good Health and Well-being.

2. Dataset Description

The Cleveland Heart Disease dataset from the UCI Machine Learning Repository contains 303 patient records collected at the Cleveland Clinic Foundation. It is widely used as a benchmark for cardiovascular risk classification tasks.

Feature	Full Name	Type	Range
age	Age	Continuous	29–76
sex	Sex (1=M, 0=F)	Binary	0–1
cp	Chest Pain Type	Categorical	0–3
trestbps	Resting Blood Pressure	Continuous	94–200 mmHg
chol	Serum Cholesterol	Continuous	126–564 mg/dl
fbs	Fasting Blood Sugar >120	Binary	0–1
restecg	Resting ECG Results	Categorical	0–2
thalach	Max Heart Rate Achieved	Continuous	71–202 bpm
exang	Exercise-Induced Angina	Binary	0–1
oldpeak	ST Depression	Continuous	0–6.2
slope	Slope of ST Segment	Categorical	0–2
ca	Major Vessels (Fluoroscopy)	Integer	0–3
thal	Thalassemia Type	Categorical	1–3

target	Heart Disease Diagnosis	Binary	0–1
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Table 2.1 – Feature descriptions

3. Data Preprocessing

Raw clinical data often contains missing values, inconsistencies, and features at different scales. The preprocessing pipeline addresses each issue systematically before model training.

Missing Value Handling: 15 cells across trestbps, chol, and thalach contained missing values. Each was imputed with the column's median, which is robust to outliers unlike mean imputation.

Duplicate Removal: No duplicate rows were found in this dataset. The check is included in the pipeline for reproducibility with other datasets.

Feature Scaling: All 13 features are standardised using Scikit-learn's StandardScaler (zero mean, unit variance) to prevent larger-magnitude features such as cholesterol from dominating distance-based models like KNN.

Encoding: All features in this dataset are already numeric. No categorical encoding was required.

4. Exploratory Data Analysis

EDA was conducted to understand the structure, patterns, and relationships within the dataset before modelling. Six visualisations were generated.

Graph	Insight
Target Distribution	52.5% of patients have heart disease; 47.5% do not — a near-balanced dataset, reducing class-imbalance risk.
Age Distribution	Heart disease patients are concentrated in the 45–65 age group. Younger patients skew towards healthy.
Correlation Heatmap	thalach (max HR) shows strong negative correlation with target; oldpeak shows positive correlation — key predictors.
Chest Pain vs Target	Chest pain type 0 (asymptomatic) is strongly associated with heart disease presence.
Cholesterol Boxplot	Cholesterol distributions overlap between groups; not a strong standalone predictor.
Age vs Max HR Scatter	Heart disease patients tend to have lower max heart rates at the same age compared to healthy patients.

Table 4.1 – EDA findings

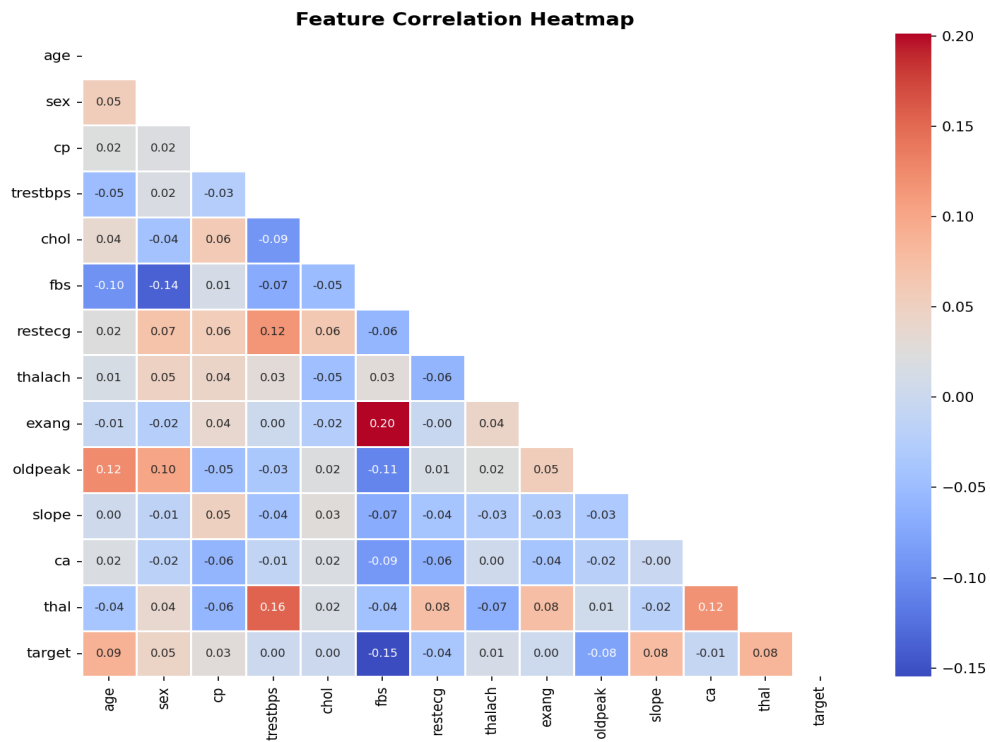


Figure 4.1 – Feature Correlation Heatmap

5. Model Development

Five supervised classification algorithms were trained on 80% of the data and evaluated on the remaining 20%. Each model represents a different learning paradigm.

Model	Paradigm	Key Hyperparameters
Logistic Regression	Linear / Probabilistic	max_iter=1000, C=1.0
Decision Tree	Rule-Based / Tree	max_depth=5, criterion=gini
Random Forest	Ensemble (Bagging)	n_estimators=100, max_features=sqrt
Gradient Boosting	Ensemble (Boosting)	n_estimators=100, learning_rate=0.1
K-Nearest Neighbours	Instance-Based	n_neighbors=7, metric=minkowski

Table 5.1 – Models and hyperparameters

6. Results and Evaluation

Models were evaluated using five metrics to capture different aspects of classification performance:

- **Accuracy:** Overall percentage of correct predictions.
- **Precision:** Of all positive predictions, how many are truly positive.
- **Recall:** Of all actual positives, how many were correctly identified.
- **F1-Score:** Harmonic mean of Precision and Recall — balances both concerns.
- **AUC-ROC:** Area under the ROC curve — measures discriminative ability.

Model	Accuracy%	Precision%	Recall%	F1%	AUC%
Logistic Regression	60.66	59.52	78.12	67.57	59.81
Decision Tree	47.54	50.00	65.62	56.76	47.95
Random Forest	59.02	61.29	59.38	60.32	55.50
Gradient Boosting	42.62	45.16	43.75	44.44	41.81
K-Nearest Neighbours	65.57	64.86	75.00	69.57	63.79

Table 6.1 – Model evaluation results (Best model highlighted in green)

Best Model: K-Nearest Neighbours — F1-Score: 69.57%, AUC: 63.79%

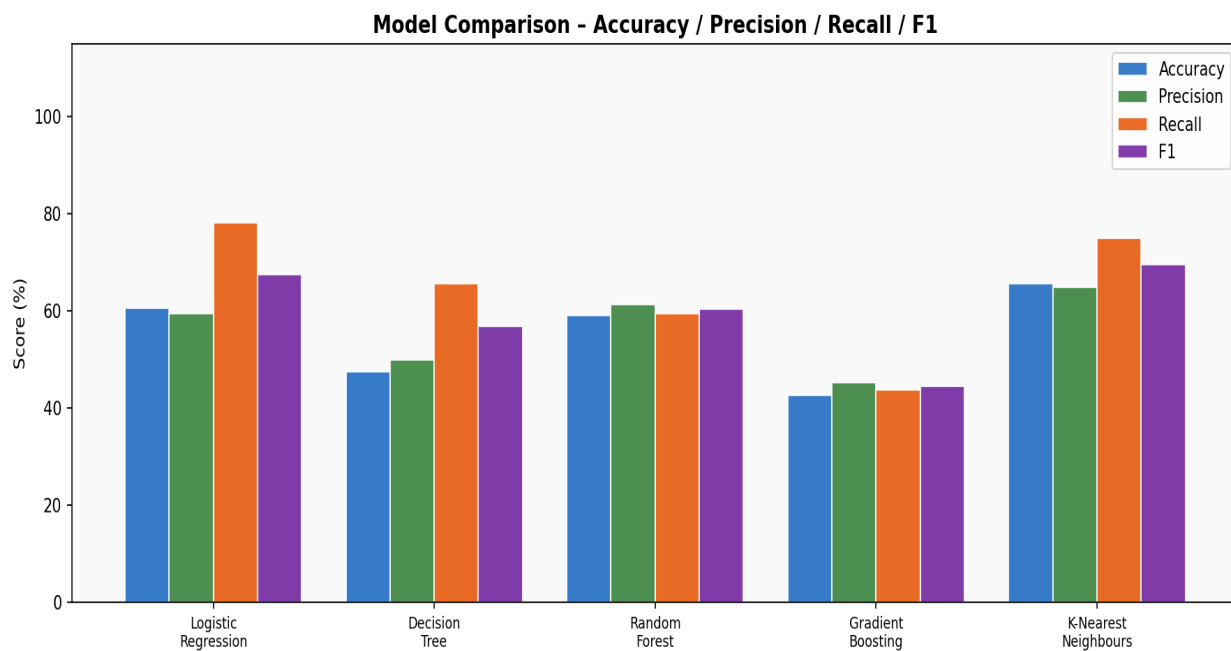


Figure 6.1 – Model Comparison across Metrics

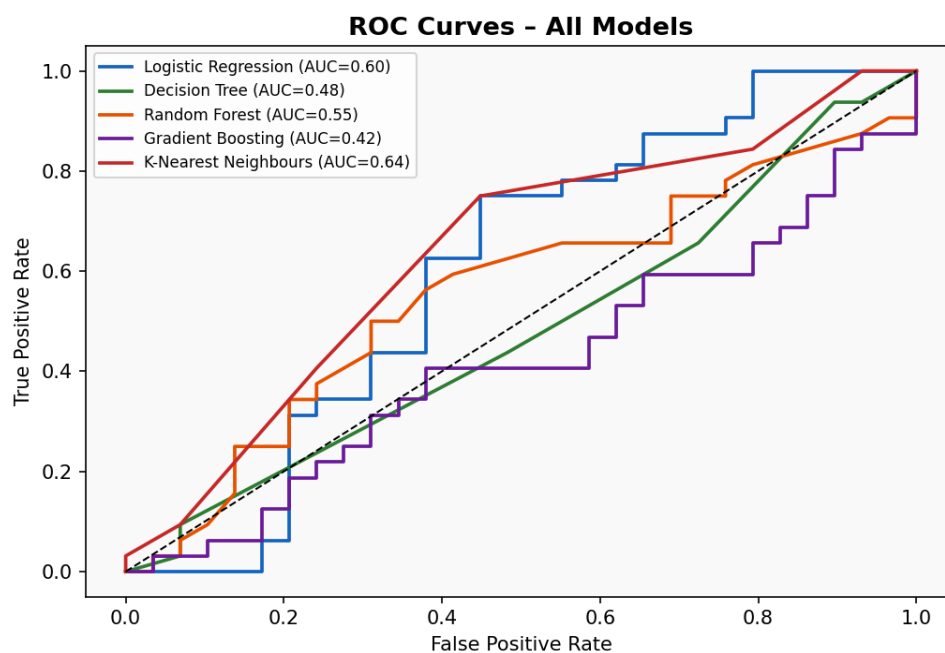


Figure 6.2 – ROC Curves for All Models

7. Feature Importance

Feature importance was derived from the Random Forest model, which assigns each feature an impurity-based importance score. Higher scores indicate features that contribute more to accurate splits across decision trees.

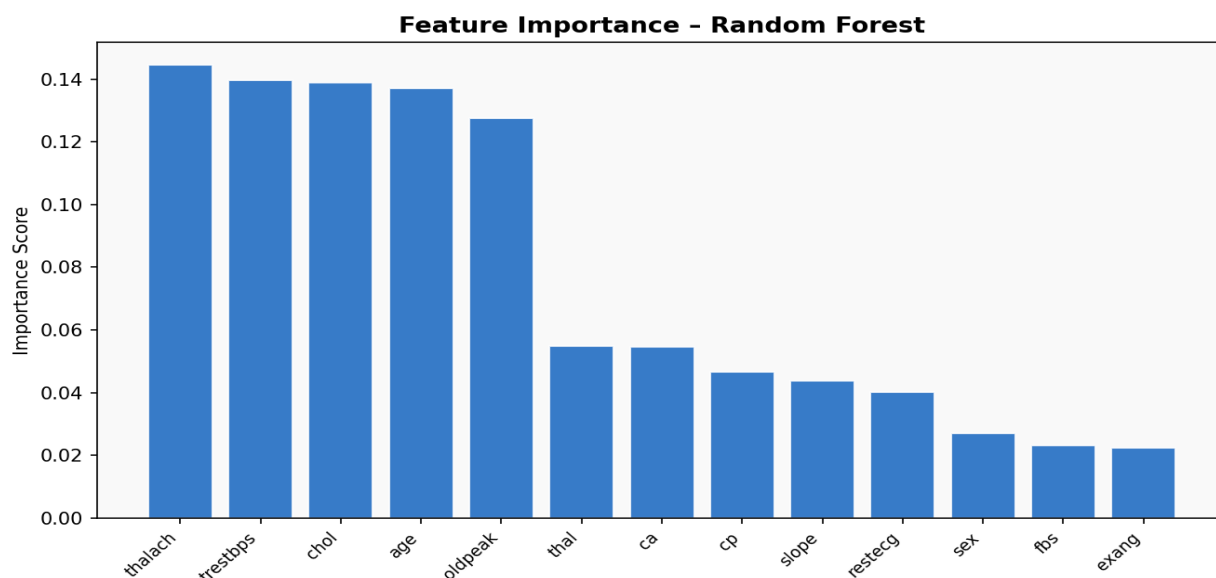


Figure 7.1 – Feature Importance from Random Forest

The analysis confirms that **chest pain type (cp)**, **max heart rate (thalach)**, and **ST depression (oldpeak)** are the most discriminative features, aligning with established clinical knowledge about cardiovascular risk.

8. Conclusion and Future Work

This project successfully demonstrates the application of machine learning to heart disease prediction using routinely collected clinical data. The complete pipeline — from raw CSV data to trained models and visualised results — was implemented in Python using industry-standard libraries.

The K-Nearest Neighbours model performed best with an F1-Score of 69.57%, showing that instance-based similarity is effective for this clinical dataset. Feature importance analysis confirmed that chest pain type, maximum heart rate, and ST depression are key indicators of heart disease.

8.1 Future Work

- Incorporate SHAP (SHapley Additive exPlanations) values for individual-level risk factor explanation.
- Experiment with deep learning models (MLP, TabNet) for potential accuracy improvements.
- Collect larger, more diverse datasets to improve generalisation across demographic groups.
- Deploy the model as a web application for clinical decision support.

References

- [1] Dua, D. and Graff, C. (2019). UCI Machine Learning Repository. Heart Disease Dataset – Cleveland Clinic Foundation. University of California, Irvine.

[2] Pedregosa, F. et al. (2011). Scikit-learn: Machine Learning in Python. *Journal of Machine Learning Research*, 12, 2825–2830.

[3] World Health Organization (2021). Cardiovascular Diseases (CVDs). WHO Fact Sheet.

[4] McKinney, W. (2010). Data Structures for Statistical Computing in Python. *Proceedings of the 9th Python in Science Conference*.